LUNA ELLIOTT

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Publications may appear under the name Luke Elliott.

RESEARCH PROFILE

I work in topological algebra, geometric group theory and semigroup theory. Author of 20 research papers (11 published and 12th accepted), including publications in Transactions of the American Mathematical Society, Advances in Mathematics, Journal of Algebra and Mathematics of Computation.

SELECTED PAPERS (10 OF 20)

See the complete list by clicking here.

- Automatic continuity, unique Polish topologies, and Zariski topologies on monoids and clones. L Elliott, J Jonušas, Z Mesyan, J Mitchell, M Morayne, Y Peresse. Transactions of the American Mathematical Society. 2023
- Sufficient conditions for a group of homeomorphisms of the Cantor set to be two-generated. C Bleak, L Elliott, J Hyde. Journal of the Institute of Mathematics of Jussieu. 2024
- A short proof of Rubin's theorem. J Belk, L Elliott, F Matucci. Israel Journal of Mathematics. 2025
- Polish topologies on endomorphism monoids of relational structures. L Elliott, J Jonušas, JD Mitchell, Y Péresse, M Pinsker. Advances in Mathematics. 2023
- E-disjunctive inverse semigroups. L Elliott, A Levine, J Mitchell. Journal of Algebra. 2025
- Topological embeddings into transformation monoids. S Bardyla, L Elliott, J Mitchell, Y Péresse. Forum Mathematicum. 2024
- A description of Aut(dVn) and Out(dVn) using transducers. L Elliott. Groups, Geometry & Dynamics. 2023
- The Diophantine problem in Thompson's group F. L Elliott, A Levine Mathematics of Computation. 2025
- Counting monogenic monoids and inverse monoids. L Elliott, A Levine, JD Mitchell. Communications in Algebra. 2023
- The Zariski Topology on Homeomorphism groups. L Elliott. Topology and its Applications. 2026

DEGREES

PhD, University of St Andrews (2021)
Thesis: On constructing topology from algebra
Masters, University of St Andrews (2018)

ACADEMIC SERVICE

Co-organiser of the Manchester Heilbronn Semigroup Seminar. Co-organiser of the European Mathematical Society's semigroup network (EMS TAG). Referee for peer-reviewed mathematics journals.

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EMPLOYMENT HISTORY

- 2024-present: University of Manchester Heilbronn Fellow. The Heilbronn Institute for Mathematical Research (HIMR), is a UK national research institute funded by GCHQ. My time is divided equally between working on fundamental mathematical research projects relating to GCHQ's long-term strategic interests (which are classified), and my own personal research (which I can publish). Fellowships are awarded competitively based on a personal research proposal, and equate to a personal grant of £200K over 3 years.
- 2022-2024: Visiting Assistant Professor, Dept. Mathematics, Binghamton University, New York State University.
- 2022: Research Fellow, University of St Andrews, Programming (in GAP) algorithms for working with the GNS rational groups, automorphisms of the shift and Thompson groups.
- 2018-2021: PhD Student at the University of St Andrews.
- 2017: Laidlaw Undergraduate Research and Leadership Program: Algorithms for commutative semigroups (awarded £4000).

TEACHING

See more teaching information by clicking here.

- 2022-2024: Lecturer for six undergraduate mathematics courses (300+ lecture hours). Responsible for course delivery, lecture preparation, office hours, assessment design, writing and marking examinations, quizzes and coursework.
- 2018-2021: Tutor, University of St Andrews. Delivered tutorials for undergraduate mathematics courses during my PhD.

SELECTED INVITED TALKS

See my complete list of 28 talks by clicking here.

- Reconstructing Topologies on Monoids: Invited speaker at ATTI (Online, University of Udine, Italy) 2026
- Thompson Monoids: Invited speaker at RIMS conference (Kyoto, Japan) 2026
- Thompson Partition Monoids and Jónsson-Tarski algebras: Invited speaker at TCA (Guimarães, Portugal) 2026
- Topologizing Endomorphism Monoids: Invited speaker at AAA (Vienna, Austria) 2026
- Simultanious Conjugacy in V: Invited speaker at LiT IV Landing in Topology (Pisa, Italy) 2026

RESEARCH FUNDING

Awarded three competitive Heilbronn Institute research grants (principal applicant once; co-applicant twice).
Grants where for the amount of:
Up to £3000 on 'E-Disjunctive inverse semigroups',
Up to £5000 on 'Use of Practical Algorithms to study inverse semigroups GAP',
Up to £3531 on 'Algorithms for automatic semigroups'.

INTERESTS

Maths, Programming, Indoor Climbing, Crochet, and Conversational Japanese.

REFERENCES

Available on request.